



* BEGIN *

**65 nm Logic and Mixed-Mode Standard
Performance Low-K Process Design Support
Manual**

(Ver. 1.6_P. 1)

P-Sub, Logic and Mixed-Mode Process

DSM NO. : G-01-LOGIC/MIXED_MODE65N-SP/LOW_K-DSM

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TECHNICAL INFORMATION CENTER
UNITED MICROELECTRONICS CORPORATION GROUP
No. 3 Li-Hsin Rd. 2, Science-Based Industrial Park, Hsin-Chu City, Taiwan, R.O.C.
Tel:+886-3-578-2258 Fax:+886-3-577-8271

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2. Technology Introduction

2.1 Scope

UMC L65SP is a 65nm generation CMOS process technology based on P-EPI/P-Sub structure with 1P10M, 1P9M2F, 1P9M2H backend metal and Al-Pad, Cu-Pad options. 1P10M is 10 layers of copper metal with Low-K(M1-M6)/FSG (M7-M10) and 1P9M2F/1P9M2H is 9 layers of copper metal with Low-K(M1-M7)/FSG(M8;M9) dielectrics. This technology is for the generic process. L65SP DSM includes EDR(G-02), TLR(G-03), **InterCap(G-4B, G-4B1, G-4B2, G-4D, G-4E, G-4E1, G-4E2, G-4F, G-4G, G-4G1, G-4G2, G-4H, G-4M, G-4M1, G-4M2, G-4N, G-4N1, G-4N2, G-4O, G-4O1, G-4O2)**, SPICE model(G-05), **Mask Tooling** (G-06) and cell rule(G-1C) documents, and the options of sub-intercap documents correspond to different metal thickness choices (1P10M, 1P9M2H and 1P9M2F with Al or Cu Pad). UMC's L65SP platform process offers three thicknesses of Al Pad, 12KÅ, 25KÅ and 36KÅ.

UMC's L65SP platform process offers four types of core devices, the SP_LVT, SP_RVT, SP_HVT and SP_SHVT. Each device type can stand alone or be integrated onto a single chip. All three devices are capable of operating at either 1.0V or 1.1V supply. In addition, three choices of thick oxide for I/O 1.8V, 2.5V and 3.3V device are also available, which only one can be chosen.

UMC's platform also offers MIMCAP (Metal Insulator Metal Capacitor) and 32.5kÅ top metal options for Mixed Signal or RF applications. For intercap, please refer to following table to choose suitable subset of G-04 documents.

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2.2 Key Process Features

P-EPI/P-Sub, CMOS Process with Deep N-well option.

193nm Lithography

Four Vt core and Single Thick Gate Devices

Dual Poly Gate with NiSi2

Up to 1P10M Cu process with Low-K (M1~M6) / FSG (M7~M10) or 1P9M2H Cu process with Low-K (M1-M7) /

FSG Dielectrics for 6X metal layers(M8;M9) Dielectrics or 1P9M2F Cu process with Low-K (M1-M7) / FSG

Dielectrics for 4X metal layers (M8;M9)

Al-Pad / Cu-Pad

Wire Bond / Flip Chip

Split Word Line based on 6T 0.62, 0.499 and 8T 1.158 μm^2 dense SRAM Bit Cell

2.3 Key Device Parameters

Device Type (*1)	LVT	RVT	HVT	SHVT	LVT	RVT	HVT	SHVT	1.8V I/O	2.5V I/O	2.5V Operated at 3.3V	2.5V Operated at 1.8V	3.3V I/O
Vcc (V)	1.0	1.0	1.0	1.0	1.1	1.1	1.1	1.1	1.8	2.5	3.3	1.8	3.3
Electrical Tox (Å)	20/21.5	20/21.5	20/21.5	20/21.5	20/21.5	20/21.5	20/21.5	20/21.5	39/41	62/64.5	62/64.5	62/64.5	72/74.5
Lmin (um)	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.18	0.24	0.45/0.38 (N/P)	0.22	0.37
Vt_sat N/P (V)	0.185/0.145	0.23/0.19	0.30/0.265	0.36/0.325	0.175/0.135	0.22/0.18	0.29/0.255	0.350/0.325	0.47/0.40	0.44/0.40	0.597/0.425	0.386/0.395	0.507/0.535
I _{dsat} N/P (uA/um)	900/425	830/390	695/330	585/265	1070/520	1005/480	860/410	730/340	600/255	600/275	582/289	403/161	612/296
Gate Delay (pSec/stage) (*2)	5.1	6.0	7.4	10.5	4.5	5.2	6.3	8.7	20.9	22.5	40.9	26.3	34.4

Note*1 Above device parameter are absolute **values**

Note*2 Estimated without interconnect loading

2.4 Key Device Design Rules

Table: Key Device Design Rules

Device Type Key Design Rules (um)	LVT (1.0V)			RVT (1.0V)			HVT (1.0V)			SHVT (1.0V)			LVT (1.1V)		
	Width	Space	Pitch	Width	Space	Pitch	Width	Space	Pitch	Width	Space	Pitch	Width	Space	Pitch
Diffusion	0.08	0.11	0.19	0.08	0.11	0.19	0.08	0.11	0.19	0.08	0.11	0.19	0.08	0.11	0.19
Inter-Well (*3)	NA	0.15	NA	NA	0.15	NA	NA	0.15	NA	NA	0.15	NA	NA	0.15	NA
Poly (for NMOS)	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19
Poly (for PMOS)	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19
Contact	0.09	0.11	0.20	0.09	0.11	0.20	0.09	0.11	0.20	0.09	0.11	0.20	0.09	0.11	0.20
Metal1	0.09	0.09	0.18	0.09	0.09	0.18	0.09	0.09	0.18	0.09	0.09	0.18	0.09	0.09	0.18
Via1 - Via5	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20
Metal2 - Metal6	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20
Via6 - Via7 (for 2X)	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40
Metal7 - Metal8 (for 2X)	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40
Via7 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Metal8 - Metal9 (for 6X)	0.56	0.56	1.12	0.56	0.56	1.12	0.56	0.56	1.12	0.56	0.56	1.12	0.56	0.56	1.12
Via8 - Via9 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Metal9 - Metal10 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Via9 (for 32.5KÅ)	0.60	0.75	1.35	0.60	0.75	1.35	0.60	0.75	1.35	0.60	0.75	1.35	0.60	0.75	1.35
Metal10 (for 32.5KÅ)	2.0	2.0	4.0	2.0	2.0	4.0	2.0	2.0	4.0	2.0	2.0	4.0	2.0	2.0	4.0

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Table: Key Device Design Rules (continued)

Device Type Key Design Rules (um)	RVT (1.1V)			HVT (1.1V)			SHVT (1.1V)			1.8V I/O			2.5V I/O		
	Width	Space	Pitch	Width	Space	Pitch	Width	Space	Pitch	Width	Space	Pitch	Width	Space	Pitch
Diffusion	0.08	0.11	0.19	0.08	0.11	0.19	0.08	0.11	0.19	0.24	0.18	0.42	0.32	0.18	0.50
Inter-Well (*3)	NA	0.15	NA	NA	0.15	NA	NA	0.15	NA	NA	0.15	NA	NA	0.15	NA
Poly (for NMOS)	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19	0.18	0.18	0.36	0.24	0.18	0.42
Poly (for PMOS)	0.06	0.13	0.19	0.06	0.13	0.19	0.06	0.13	0.19	0.18	0.18	0.36	0.24	0.18	0.42
Contact	0.09	0.11	0.20	0.09	0.11	0.20	0.09	0.11	0.20	0.09	0.11	0.20	0.09	0.11	0.20
Metal1	0.09	0.09	0.18	0.09	0.09	0.18	0.09	0.09	0.18	0.09	0.09	0.18	0.09	0.09	0.18
Via1 - Via5 (for 1X)	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20
Metal2 - Metal6 (for 1X)	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20
Via6 - Via7 (for 2X)	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40
Metal7 - Metal8 (for 2X)	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40
Via7 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Metal8 - Metal9 (for 6X)	0.56	0.56	1.12	0.56	0.56	1.12	0.56	0.56	1.12	0.56	0.56	1.12	0.56	0.56	1.12
Via8 - Via9 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Metal9 - Metal10 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Via9 (for 32.5KÅ)	0.60	0.75	1.35	0.60	0.75	1.35	0.60	0.75	1.35	0.60	0.75	1.35	0.60	0.75	1.35
Metal10 (for 32.5KÅ)	2.0	2.0	4.0	2.0	2.0	4.0	2.0	2.0	4.0	2.0	2.0	4.0	2.0	2.0	4.0

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Table: Key Device Design Rules (continued)

Device Type Key Design Rules (um)	2.5V Operated at 3.3V			2.5V Operated at 1.8V			3.3V I/O		
	Width	Space	Pitch	Width	Space	Pitch	Width	Space	Pitch
Diffusion	0.32	0.18	0.50	0.32	0.18	0.50	0.32	0.18	0.50
Inter-Well (*3)	NA	0.15	NA	NA	0.15	NA	NA	0.15	NA
Poly (for NMOS)	0.45	0.18	0.63	0.22	0.18	0.40	0.37	0.18	0.55
Poly (for PMOS)	0.38	0.18	0.56	0.22	0.18	0.40	0.37	0.18	0.55
Contact	0.09	0.11	0.20	0.09	0.11	0.20	0.09	0.11	0.20
Metal1	0.09	0.09	0.18	0.09	0.09	0.18	0.09	0.09	0.18
Via1 - Via5 (for 1X)	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20
Metal2 - Metal6(for 1X)	0.10	0.10	0.20	0.10	0.10	0.20	0.10	0.10	0.20
Via6 - Via7 (for 2X)	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40
Metal7 - Metal8 (for 2X)	0.20	0.20	0.40	0.20	0.20	0.40	0.20	0.20	0.40
Via7 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Metal8 - Metal9 (for 6X)	0.56	0.56	1.12	0.56	0.56	1.12	0.56	0.56	1.12
Via8 - Via9 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Metal9 - Metal10 (for 4X)	0.40	0.40	0.80	0.40	0.40	0.80	0.40	0.40	0.80
Via9 (for 32.5KÅ)	0.60	0.75	1.35	0.60	0.75	1.35	0.60	0.75	1.35
Metal10 (for 32.5KÅ)	2.0	2.0	4.0	2.0	2.0	4.0	2.0	2.0	4.0

Note*3 Inter-well defines the minimum spacing of N_WELL to N+ DIFFUSION

* END *

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